Fuel Consumption and Route Comparison

Problem statement

Compare several vehicle options using route distance, mileage, fuel price, and reserve requirements; identify the lowest-cost feasible vehicle.

Assumptions and rules

• Fuel required = distance / mileage.

• Required fuel including reserve = trip fuel × (1 + reserve percentage / 100).

• A vehicle is feasible if its available fuel is at least the required fuel including reserve.

• Journey cost is calculated using required fuel including reserve multiplied by fuel price per litre.

Algorithm

1. Read route distance, reserve percentage, and vehicle records.

2. For each vehicle, validate mileage and available fuel.

3. Calculate trip fuel and reserve-inclusive fuel requirement.

4. Calculate journey cost using the fuel price.

5. Mark the vehicle feasible if available fuel meets the requirement.

6. Track the feasible vehicle with the lowest cost.

7. Display calculations and the best feasible option.

Pseudocode

```
START\nINPUT distance, reserve_percent, vehicles\nlowest_cost = large number; best_vehicle = NONE\nFOR each vehicle\n  UNPAC
```

Flowchart

```
[Start] -> [Read distance and vehicle data]\n   |\n[Calculate trip fuel and reserve]\n   |\n[Calculate journey cost]\n   |\
```

Python solution

The program uses beginner-level Python constructs from the assigned syllabus: variables, data types, operators, conditions, loops, strings, lists, tuples, and unpacking where appropriate.

```
distance = 350
reserve_percent = 10
vehicles = [
    ("Car A", 15, 105, 30),
    ("Car B", 20, 105, 25),
    ("Car C", 12, 100, 40)
]
# Each record: name, mileage km/litre, fuel price, fuel available

lowest_cost = 0
best_vehicle = ""
found = False

for vehicle in vehicles:
    name, mileage, fuel_price, available_fuel = vehicle

    if mileage <= 0:
        print(name, "Invalid mileage")
    else:
        trip_fuel = distance / mileage
        required_fuel = trip_fuel * (1 + reserve_percent / 100)
        cost = required_fuel * fuel_price

        if available_fuel >= required_fuel:
            status = "FEASIBLE"
```

```
            if found == False or cost < lowest_cost:
                lowest_cost = cost
                best_vehicle = name
                found = True
        else:
            status = "NOT FEASIBLE"

        print(name, "Trip fuel:", round(trip_fuel, 2),
              "With reserve:", round(required_fuel, 2),
              "Cost:", round(cost, 2), status)

if found:
    print("Lowest-cost feasible vehicle:", best_vehicle)
    print("Estimated cost:", round(lowest_cost, 2))
else:
    print("No vehicle has enough fuel")
```

Sample output

```
Car A Trip fuel: 23.33 With reserve: 25.67 Cost: 2695.0 FEASIBLE\nCar B Trip fuel: 17.5 With reserve: 19.25 Cost: 2021.25 F
```

Explanation / observations

The comparison assumes a single fuel price per vehicle and treats the reserve fuel as part of the planned fuel requirement and cost.

Conclusion

This solution models the stated case using a clear sequence of input, processing, decisions, repetition, and output. The assumptions and sample data can be changed to test other cases.